Target-Side Semantic Anchoring for Low-Resource Neural Machine Translation

Research Proposal

Abstract

Low-resource neural machine translation (NMT) is commonly optimized as conditional token prediction even when the target language is represented by a substantially stronger monolingual or multilingual pretrained model. We propose TSSA, a general framework that treats the rich target language as a training-time semantic anchor rather than only an output vocabulary. A frozen target-side teacher supplies token- and sentence-level representations; the source encoder is shaped toward that space through confidence-weighted structural anchoring and in-batch semantic priming; and lightweight head-wise routing learns which cross-attention channels should transmit anchor-consistent evidence. The core method adds no target-reference input at inference. We study Bahnaric–Vietnamese, Tay–Vietnamese, and Rhade–Vietnamese translation with BARTpho and multilingual sequence-to-sequence backbones. The proposal tests whether asymmetric anchoring improves sample efficiency, adequacy, alignment quality, robustness, and hallucination behavior at acceptable computational cost. Wake–Sleep Semantic Consolidation is retained as an optional extension rather than part of the core claim.

1 Motivation and Research Question

Parallel corpora for minoritized languages are limited, noisy, and lexically sparse. Standard NMT asks a weakly pretrained source representation to support a difficult autoregressive objective, although the target language may already have a strong semantic space. Our central question is: *Can a rich target-side model act as a latent teacher that organizes source representations before and during translation?* We hypothesize that the benefit is asymmetric and largest at the lowest data scales.

A central practical obstacle is that two common low-resource strategies, back-translation and pivot translation, move the problem rather than remove it. Back-translation requires a reverse target-to-source model that is often weakest exactly when the source language is severely under-resourced; errors from that reverse model become synthetic-source noise. Pivot translation requires one or two usable auxiliary translation directions and may compound lexical loss, reordering errors, and domain mismatch across stages. Both strategies therefore introduce an intermediate-model bottleneck and additional decoding cost. TSSA is motivated by a stricter alternative: exploit the rich target model directly as a frozen semantic teacher during training, without requiring a competent reverse model, a pivot language, or an intermediate translation at inference.

The proposal is cognitively motivated but not neuroanatomically identified. Statistical distributional learning motivates repeated co-occurrence evidence, while bilingual-memory studies motivate asymmetric lexical-to-conceptual access and target-side priming [7, 15, 23, 24]. Work on second-language neuroplasticity and predictive language processing supports treating learning, prediction, and repair as interacting constraints, without claiming that Transformer components correspond one-to-one to brain regions [18, 21]. Classical critiques of purely associative accounts also caution that frequency alone cannot explain structured language behavior [6]. Thus, cognition is inspiration; all scientific claims are stated and tested at the representation and behavior levels.

Primary hypothesis. Given source language S, rich target language T, parallel pair $(\boldsymbol{x}, \boldsymbol{y})$, and a strong target encoder, target-side semantic anchoring will improve translation most when parallel supervision is scarce, because the target teacher supplies a lower-variance semantic organization for the source encoder.

Falsifiable predictions. (i) Gains decrease as parallel data increases; (ii) true target anchors outperform shuffled, unrelated-language, and random-teacher controls; (iii) removing the identified anchor-sensitive heads causes a larger adequacy loss than removing matched random heads; and (iv) improvements correlate with better independently measured alignment and lexical adequacy, not only surface overlap.

2 Positioning

Attention is useful but is not automatically a word alignment model [11, 19, 22]. Alignment can be induced more accurately by shifting extraction time, adding explicit objectives, or learning end-to-end aligners [5, 10, 31]. Recent work further uses alignment as a preference signal for reducing omission and hallucination, and shows that multilingual MT encoders can be strong aligners in their own right [9, 30]. TSSA therefore does not equate cross-attention weights with gold links. It uses an external or held-out alignment posterior as a training scaffold and evaluates alignment separately.

Cross-lingual mapping has progressed from approximately linear lexical transformations to robust self-learning, multilingual pretraining, and multiple local subspaces [1, 13, 16, 24]. Bilingual lexicons can also be induced through unsupervised MT, mined bitext, and word alignment [2, 27]. Surveys emphasize that alignment may mean geometric similarity, task transferability, or token correspondence and that these notions should not be conflated [12]. Multilingual NMT similarly benefits from separating semantic sharing from target-language-specific features [3]. BART, MASS, and mBART establish strong denoising or masked-sequence pretraining baselines for complete encoder–decoder models [17, 20, 28], while extensible mBART fine-tuning supports multilingual transfer [29].

Back-translation is an influential use of target-side monolingual data, but it first translates target sentences into the low-resource source language and consequently depends on a target-to-source generator [26]. Pivot systems similarly route translation through an auxiliary language or direction. These methods are valuable when the intermediate models are reliable, yet they are structurally disadvantaged in the extreme low-resource regime studied here: the reverse or pivot system can become the quality ceiling and can propagate its errors into training or inference. We therefore treat back-translation and pivoting as related paradigms, not experimental baselines for the core comparison. Our distinction is to transfer target-side semantic structure directly into the source representation during supervised training, making alignment explicitly *target-anchored, asymmetric, confidence-weighted, and coupled to efficient cross-attention routing* without an intermediate translation model.

3 Proposed Method

3.1 Setting and Baseline Objective

Let $\boldsymbol{x} = (x_1, \ldots, x_m)$ and $\boldsymbol{y} = (y_1, \ldots, y_n)$. The trainable translation model contains source encoder E_S, decoder D, and output parameters θ. A frozen target teacher E_T is initialized from a strong target-language or multilingual sequence-to-sequence model. The standard teacher-forced NMT loss is

$$\mathcal{L}_{\mathrm{MT}}(\theta) = -\frac{1}{n}\sum_{t=1}^{n} \log p_\theta(y_t \mid y_{<t}, \boldsymbol{x}). \tag{1}$$

The target teacher processes $\boldsymbol{y}$ *only during training.* It is never consulted at inference, preventing target-reference leakage.

3.2 Confidence-Weighted Structural Anchoring

For training pairs, an offline aligner produces posterior matrix $A \in [0,1]^{m\times n}$, where A_{ij} is confidence that source token x_i corresponds to target token y_j. Rows are normalized over supported links; unaligned rows are masked. This posterior can be produced by a strong multilingual encoder aligner and symmetrized where bidirectional scores are available. The source and teacher states are $\boldsymbol{h}_i^S = E_S(\boldsymbol{x})_i$ and $\boldsymbol{h}_j^T = E_T(\boldsymbol{y})_j$. The target barycenter for source position i is

$$\bar{\boldsymbol{h}}_i^T = \sum_{j=1}^{n} \widetilde{A}_{ij}\boldsymbol{h}_j^T, \qquad \widetilde{A}_{ij} = \frac{A_{ij}}{\sum_{k=1}^{n} A_{ik} + \varepsilon}. \tag{2}$$

With learned projections P_S, P_T into the same dimension and confidence $c_i = \min(1, \sum_j A_{ij})$, the structural loss is

$$\mathcal{L}_{\mathrm{struct}} = \frac{1}{\sum_i c_i + \varepsilon}\sum_{i=1}^{m} c_i \left\| \frac{P_S\boldsymbol{h}_i^S}{\left\|P_S\boldsymbol{h}_i^S\right\|_2 + \varepsilon} - \mathrm{sg}\left(\frac{P_T\bar{\boldsymbol{h}}_i^T}{\left\|P_T\bar{\boldsymbol{h}}_i^T\right\|_2 + \varepsilon}\right)\right\|_2^2. \tag{3}$$

The stop-gradient operator $\mathrm{sg}(\cdot)$ and frozen E_T make the target space an anchor rather than a moving co-adaptation target. Confidence weighting limits damage from uncertain word links. This term aligns contextual token geometry; it does not claim that the external links are gold linguistic boundaries.

3.3 Sentence-Level Semantic Priming

Token links may be missing for rare forms, spelling variants, or divergent word order. We therefore add a global contrastive objective. Let $\boldsymbol{z}_b^S$ and $\boldsymbol{z}_b^T$ be ℓ_2-normalized, mask-aware mean pools of projected states for batch item b. For batch size B and temperature $\tau > 0$,

$$\mathcal{L}_{\mathrm{prime}} = -\frac{1}{B}\sum_{b=1}^{B} \log \frac{\exp((\boldsymbol{z}_b^S)^\top \mathrm{sg}(\boldsymbol{z}_b^T)/\tau)}{\sum_{k=1}^{B}\exp((\boldsymbol{z}_b^S)^\top \mathrm{sg}(\boldsymbol{z}_k^T)/\tau)}. \tag{4}$$

Other target sentences in the minibatch are negatives. Duplicate or semantically near-duplicate targets will be masked when metadata or similarity screening identifies likely false negatives. This term shapes sentence semantics while $\mathcal{L}_{\mathrm{struct}}$ supplies local correspondence.

3.4 Anchor-Consistent Head-Wise Routing

Edge-wise gates over all source–target token pairs would add substantial cost and could overfit alignment noise. Instead, each decoder cross-attention head receives a scalar gate. At decoder layer ℓ, head h, and target step t, let $\boldsymbol{o}_{\ell ht}$ be the ordinary head output, $\boldsymbol{q}_{\ell ht}$ its query, and $\boldsymbol{s}$ a pooled source summary. A small router predicts

$$g_{\ell ht} = \sigma(\mathrm{MLP}_{\ell h}([\boldsymbol{q}_{\ell ht}; \boldsymbol{s}; \boldsymbol{q}_{\ell ht} \odot \boldsymbol{s}])), \qquad \widetilde{\boldsymbol{o}}_{\ell ht} = g_{\ell ht}\boldsymbol{o}_{\ell ht}. \tag{5}$$

The router uses only quantities available at inference. During teacher-forced training, a reliability target is derived from agreement between the projected head output and the frozen teacher state at the same target step:

$$r_{\ell ht} = \frac{1 + \cos(R_{\ell h}\boldsymbol{o}_{\ell ht},\, \mathrm{sg}(P_T \boldsymbol{h}_t^T))}{2} \in [0, 1], \tag{6}$$

where $R_{\ell h}$ is a learned projection. The routing supervision is soft binary cross-entropy,

$$\mathcal{L}_{\mathrm{route}} = -\frac{1}{N} \sum_{\ell,h,t} \left[r_{\ell ht} \log g_{\ell ht} + (1 - r_{\ell ht}) \log(1 - g_{\ell ht}) \right]. \tag{7}$$

The reliability target is detached when optimizing the gate, preventing a trivial solution in which the representation changes merely to make its own label easier. The translation loss still trains the gated path end to end. A warm-up period with $g = 1$ will stabilize the baseline before routing supervision is activated.

3.5 Unified Objective and Training Schedule

The core objective is

$$\boxed{\mathcal{L} = \mathcal{L}_{\mathrm{MT}} + \lambda_1 \mathcal{L}_{\mathrm{struct}} + \lambda_2 \mathcal{L}_{\mathrm{prime}} + \lambda_3 \mathcal{L}_{\mathrm{route}}.} \tag{8}$$

All terms are computed on genuine parallel minibatches. We will use a short translation-only warm-up, linearly ramp λ_1 and λ_2, and activate routing after the source space begins to align. Hyperparameters are selected only on validation data. Gradient norms and per-loss scales will be logged; if one auxiliary term dominates, coefficients will be normalized by a fixed initial-scale estimate rather than tuned on test performance.

Optional extension. Wake–Sleep Semantic Consolidation generates synthetic source–target pairs after the core model converges, filters them using translation confidence, anchor agreement, and structural coverage, then continues training with a capped synthetic-data ratio. Because generation can dominate runtime and introduces additional confounds, it will be reported as TSSA+Sleep and will not be required to validate the core hypothesis.

4 Experimental Plan

4.1 Data and Documentation

We will evaluate three community-relevant $X \rightarrow$Vietnamese directions: Bahnaric–Vietnamese, Tay–Vietnamese, and Rhade–Vietnamese. The Bahnaric corpus was collected through fieldwork in local communities; the Tay corpus was developed with a Tay-language lecturer in Cao Bang; and the Rhade corpus was crawled from Bible text and electronic articles from Vietnamese ethnic-minority media. Dataset cards, licensing status, consent/provenance, orthographic conventions, deduplication, and train/dev/test construction will be audited before release or experimentation. The current repositories are:

- `https://huggingface.co/datasets/FiveC/bahnaric_vietnamese`
- `https://huggingface.co/datasets/HeyDunaX/tay-vietnamese-nmt`
- `https://huggingface.co/datasets/NIRVLab/rhade-vietnamese-mt`

Splits will be document- or source-aware whenever identifiers permit, with exact and near-duplicate removal across splits. Unicode normalization, punctuation, script variation, spelling variants, and language identification will be reported rather than silently standardized. Evaluation references will be manually inspected by speakers or qualified annotators where feasible.

4.2 Models and Initialization

The principal target teacher is BARTpho. Translation backbones will include: (i) a six-layer encoder–decoder Transformer trained from scratch; (ii) BARTpho converted into an $X \rightarrow$Vietnamese model by replacing or extending the source embedding/tokenizer interface while retaining the pretrained Vietnamese decoder; and (iii) mBART-style multilingual encoder–decoders. BART is a denoising sequence-to-sequence model rather than a conventional encoder-only MLM, and mBART extends this objective to multilingual full-model pretraining [17, 20]. Accordingly, we use the precise term *denoising language-model adaptation* below; an encoder-only masked-token objective is included only where the architecture supports it. MASS is included as the masked-sequence encoder–decoder analogue [28].

For each backbone, tokenizer/vocabulary policy, embedding initialization, number of trainable parameters, optimizer, and update budget will be matched. When a source vocabulary must be added, all compared methods receive the same vocabulary and initialization. The frozen target teacher is not counted as an inference-time component, but its training FLOPs and cached-state storage are reported.

4.3 Baseline Families

A credible comparison must separate target semantic anchoring from ordinary pretraining, strong low-resource tuning, explicit word alignment, and additional compute. We therefore organize the baselines into five families.

B0: translation-only controls. B0.1 Transformer-from-scratch uses only parallel maximum likelihood. **B0.2 Backbone fine-tuning** fine-tunes BARTpho or mBART directly on $X \to$Vietnamese. **B0.3 Low-resource optimized NMT** applies the same vocabulary size, dropout, label smoothing, batch construction, early stopping, and hyperparameter search budget as TSSA, following evidence that careful adaptation materially changes low-resource conclusions [25]. These are the primary no-auxiliary-loss references.

B1: masked/denoising language-model adaptation on the same backbone. Before translation fine-tuning, **B1.1 Target-DAE** continues BARTpho/mBART span-infilling denoising on Vietnamese monolingual text; **B1.2 Source-DAE** uses available unlabeled X text; and **B1.3 Bilingual-DAE** alternates source and target denoising batches. **B1.4 MASS-style adaptation** masks a contiguous segment on the encoder input and reconstructs it autoregressively with the decoder. Where an encoder-only MLM head is technically valid, **B1.5 Source-MLM** masks source tokens and updates the encoder before NMT fine-tuning. These controls test whether gains arise simply from more monolingual exposure or continued self-supervision. BART, mBART, and MASS supply the canonical objectives [17, 20, 28]. Every condition uses the same monolingual tokens and additional update count as closely as possible.

B2: direct transfer baselines without reverse or pivot models. B2.1 Forward self-training translates unlabeled source sentences directly into Vietnamese and retrains on confidence-filtered pseudo-pairs. **B2.2 Target-side sequence distillation** trains the student on outputs from the strongest direct $X \to$Vietnamese teacher under the same decoding policy. **B2.3 Multilingual transfer** jointly trains all three direct $X \to$Vietnamese directions with language tags. **B2.4 Target-language adapter transfer** initializes or regularizes the decoder from the rich Vietnamese model without explicit cross-lingual anchoring. No baseline in this family may use a Vietnamese$\to X$ reverse model or an auxiliary pivot language. These comparisons determine whether TSSA improves beyond additional direct pseudo-labels, smoother targets, decoder transfer, or multilingual parameter sharing while preserving the no-intermediate-model setting.

B3: alignment-aware sequence-to-sequence baselines. B3.1 Attention extraction uses standard cross-attention as a soft alignment diagnostic, while recognizing that attention need not equal alignment [11, 19, 22]. **B3.2 Shift-Att/Shift-AET** follows alignment induction at the decoder-input shift proposed by Chen et al. [5]. **B3.3 Joint alignment and translation** adds an alignment head/objective following Garg et al. [10]. **B3.4 End-to-end neural alignment** follows the alignment-learning formulation of Zenkel et al. [31]. **B3.5 Mask-Align supervision** obtains links by masked target reconstruction with full target context [4]. **B3.6 Alignment-as-preference**, if compute permits, creates chosen/rejected translations from alignment coverage and optimizes a preference objective following Wu et al. [30]. All methods will consume the same bitext and will be reimplemented on the same backbone whenever the original architecture allows; otherwise they are labeled as reference systems rather than perfectly matched ablations.

B4: external alignment and representation baselines. The alignment posterior used by TSSA will be instantiated separately with **SimAlign**, **AWESOME-align**, **Mask-Align**, and **TransAlign**. SimAlign extracts links from multilingual embeddings without parallel aligner training; AWESOME-align fine-tunes multilingual contextual representations on bitext; Mask-Align uses target masking; and TransAlign uses multilingual MT encoder states [4, 8, 9, 14]. These are both alternative posterior generators and standalone alignment references. Statistical fast-align/eflomal links will be included as low-cost traditional checks where scripts and licenses permit. The main TSSA result must remain positive with at least two reasonable posterior generators, otherwise the claim will be scoped to a particular aligner.

Table 1: Controls required to identify target-side anchoring.

Control	Purpose
Shuffled target anchors	Tests whether gains require pair-specific semantics.
Random/frozen teacher	Separates semantic knowledge from regularization and capacity.
Unrelated rich-language teacher	Tests target specificity.
No $\mathcal{L}_{\text{struct}}$, no $\mathcal{L}_{\text{prime}}$, no router	Attributes gains to components and interactions.
Random matched heads ablated	Tests causal specificity of anchor-sensitive routing.
Equal-parameter adapter baseline	Controls for additional trainable parameters.

B5: objective-matched and compute-matched controls. **B5.1 Sentence contrastive only** uses $\mathcal{L}_{\text{prime}}$ without a frozen target teacher by contrasting trainable source/target branches. **B5.2 Token alignment only** uses $\mathcal{L}_{\text{struct}}$ without sentence priming or routing. **B5.3 Equal-parameter adapters** adds the same number of trainable parameters as the router. **B5.4 Extra-update control** trains the ordinary NMT objective for the same wall-clock or optimizer steps used by TSSA. **B5.5 Shuffled, random, and unrelated-language anchors** test pair specificity and semantic relevance. Together these controls distinguish target anchoring from generic regularization, capacity, and compute.

4.4 Resource Regimes and Controlled Training

For each language, nested training subsets will be sampled at 1%, 5%, 10%, 25%, 50%, and 100% of the deduplicated training data, stratified by document/domain where possible. The main experimental matrix deliberately excludes back-translation and pivot translation because both require an intermediate translation model whose quality and compute would confound the proposed direct target-anchor mechanism. The same subset IDs will be reused across all methods. Development and test sets remain fixed and are never used to construct alignments, teacher caches, synthetic data, vocabulary statistics, or early-stopping signals.

Each primary condition will use at least three seeds, with five seeds at the smallest two budgets if variance is high. Optimization will use AdamW, inverse-square-root or linear scheduling selected on development data, gradient clipping, mixed precision, dynamic token batching, and early stopping on a preregistered validation criterion. The search space for learning rate, dropout, label smoothing, warm-up, beam size, length penalty, and $\lambda_1, \lambda_2, \lambda_3$ will be declared in advance. Baselines receive the same number of tuning trials. Checkpoints will be selected without test access, and all metric computation will be performed on detokenized outputs from the same decoding script.

4.5 Factorial Ablations

The principal ablation is a 2^3 factorial design over structural anchoring, sentence priming, and routing, including the translation-only origin and the full method. A second analysis varies teacher status (frozen, exponential-moving-average, trainable), teacher layer, alignment confidence threshold, and cached versus online teacher states. Interaction effects are important: the framework predicts that routing is most useful after token and sentence anchoring have formed a stable source-to-target geometry. Wake–Sleep is evaluated only after the full core model and is compared with direct forward self-training under the same number of retained pseudo-pairs; neither condition uses a reverse or pivot model.

4.6 Evaluation Protocol and Statistical Analysis

Primary translation metrics are sacreBLEU, chrF++, and COMET, with exact versions, signatures, tokenization, reference handling, and COMET checkpoint reported. BLEU captures corpus-level n-gram precision, chrF++ is more tolerant of morphological and orthographic variation, and COMET supplies learned adequacy-oriented evaluation. The confirmatory endpoint is macro-average COMET across the three language directions at each resource budget; language-specific BLEU and chrF++ remain co-primary descriptive outcomes. We will additionally report length ratio, source-copy rate, number and named-entity preservation, omission/addition rate, and lexical accuracy on curated frequent/rare/content-word bins.

Alignment evaluation will not reuse the posterior that trained the method as if it were gold. A small manually aligned subset will be double-annotated with adjudication, enabling alignment error rate, precision, recall, and F1. We will also report symmetrized source-to-target retrieval and alignment coverage, but label these as proxy metrics. Hallucination analysis will distinguish detached additions, mistranslation, repetition, and legitimate explicitation.

Human evaluation will use blinded, randomized pairwise adequacy judgments on a stratified sample emphasizing rare content words, culturally specific concepts, long sentences, spelling variants, and divergent word order. Fluency will be judged separately so that target-model strength cannot hide adequacy loss. Annotator instructions, qualification, compensation, disagreement, and adjudication will be documented.

Statistical testing will operate on paired sentence-level outputs. We will report paired bootstrap 95% confidence intervals for BLEU/chrF++/COMET differences, approximate randomization where assumptions are appropriate, and a hierarchical model or mixed-effects analysis across languages, resource levels, and seeds. Confirmatory comparisons against B0.3, the strongest B1 language-model adaptation, the strongest B2 transfer baseline, and the strongest B3 alignment baseline will be corrected for multiplicity. Effect sizes and uncertainty will accompany p-values; a gain observed for only one seed or one language will not support the general claim.

4.7 Reproducibility and Reporting

Every experiment will record dataset revision, split checksum, tokenizer checksum, model revision, seed, trainable parameter count, maximum tokens per batch, gradient accumulation, effective batch size, optimizer updates, best checkpoint step, GPU type, peak memory, training time, and decoding configuration. A single evaluation container will generate all detokenized outputs and metrics. The released result matrix will distinguish prespecified primary experiments from exploratory variants. Failure runs and hyperparameter trials will be retained to prevent selective reporting.

4.8 Efficiency and Acceptance Criterion

The frozen teacher adds a training-only forward pass; $\mathcal{L}_{\text{prime}}$ adds a batch similarity matrix; token anchoring adds projected distances on supported links; and head-wise routing adds small MLPs and scalar multiplications. No teacher is needed for decoding. We will report wall-clock time, tokens/s, peak memory, trainable parameters, total FLOPs when measurable, convergence steps, and inference latency. The engineering target is no more than 20% core-training wall-clock overhead after caching teacher states and offline alignment posteriors. If overhead exceeds this target, teacher layers will be selectively cached or distilled, token anchoring will be sampled, and routing will remain head-wise. TSSA+Sleep will have a separate compute ledger.

5 Mechanistic and Robustness Validation

We will identify anchor-sensitive layers and heads using gate values, gradient-based attribution, representation similarity, and causal ablation. The central causal test removes the highest-scoring anchor heads, matched random heads, and lowest-scoring heads under equal counts. A valid mechanism should show a selective loss in COMET, lexical adequacy, and alignment quality after anchor-head ablation, while unrelated fluency changes remain smaller. Representation analyses will compare source–teacher retrieval, neighborhood stability, and token correspondence across layers.

Robustness tests will introduce controlled spelling variation, diacritic deletion, word-order perturbation, and rare-word substitutions. The goal is not invariance to meaning-changing noise; it is graceful degradation when orthographic evidence is weak but contextual target semantics remain informative. We will also test domain shift and calibration. Failure cases will be categorized into wrong anchor, over-regularization toward target-language syntax, noisy alignment, untranslated copying, and culturally specific non-equivalence.

6 Expected Contributions, Risks, and Deliverables

The expected contribution is a general principle: a rich target language can serve as an asymmetric semantic scaffold for learning a low-resource source-to-target mapping. Methodologically, the work contributes confidence-weighted token anchoring, sentence-level semantic priming, and inference-compatible head-wise routing in one objective. Empirically, it contributes controlled evidence across three low-resource Vietnamese-target directions and multiple data scales. Scientifically, it tests when target anchoring helps, when it disappears, and whether its effect is causally localized.

Key risks are noisy word alignment, teacher bias, target-syntax overfitting, false contrastive negatives, and data contamination. Each risk has a direct diagnostic or control. A negative result remains informative if shuffled and true anchors behave similarly, if gains vanish after equal-compute controls, or if target anchoring improves fluency but harms adequacy. Deliverables will include cleaned split manifests where licensing permits, preprocessing and evaluation scripts, trained configurations, compute logs, alignment diagnostics, and a reproducible ablation matrix.

7 Milestones

1. Data audit, split verification, vanilla/BARTpho/mBART baselines, and fixed evaluation pipeline.
2. Frozen target teacher, cached representations, external alignment posterior, and $\mathcal{L}_{\text{struct}}$.
3. In-batch priming, routing warm-up, efficiency profiling, and hyperparameter selection on validation data.
4. Full multi-seed, multi-budget experiments; human and mechanistic evaluation; robustness and controls.

5. Optional TSSA+Sleep extension only after the core hypothesis is resolved.

References

[1] Mikel Artetxe, Gorka Labaka, and Eneko Agirre. A robust self-learning method for fully unsupervised cross-lingual mappings of word embeddings. In *Proceedings of ACL*, pages 789–798, 2018.

[2] Mikel Artetxe, Gorka Labaka, and Eneko Agirre. Bilingual lexicon induction through unsupervised machine translation. In *Proceedings of ACL*, pages 5002–5007, 2019.

[3] Mengyu Bu, Shuhao Gu, and Yang Feng. Improving multilingual neural machine translation by utilizing semantic and linguistic features. In *Findings of ACL*, pages 10410–10423, 2024.

[4] Chi Chen, Maosong Sun, and Yang Liu. Mask-align: Self-supervised neural word alignment. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, pages 4781–4791, 2021.

[5] Yun Chen, Yang Liu, Guanhua Chen, Xin Jiang, and Qun Liu. Accurate word alignment induction from neural machine translation. In *Proceedings of EMNLP*, pages 566–576, 2020.

[6] Noam Chomsky. A review of B. F. Skinner's verbal behavior. Language, 35(1), 26–58, 1959.

[7] Ton Dijkstra and Walter J. B. Van Heuven. The architecture of the bilingual word recognition system: From identification to decision. *Bilingualism: Language and Cognition*, 5(3):175–197, 2002.

[8] Zi-Yi Dou and Graham Neubig. Word alignment by fine-tuning embeddings on parallel corpora. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2112–2128, 2021.

[9] Benedikt Ebing, Christian Goldschmied, and Goran Glavaš. TransAlign: Machine translation encoders are strong word aligners, too. In *Findings of EMNLP*, pages 20736–20749, 2025. doi: 10.18653/v1/2025.findings-emnlp.1129.

[10] Sarthak Garg, Stephan Peitz, Udhyakumar Nallasamy, and Matthias Paulik. Jointly learning to align and translate with transformer models. In *Proceedings of EMNLP-IJCNLP*, pages 4453–4462, 2019.

[11] Hamidreza Ghader and Christof Monz. What does attention in neural machine translation pay attention to? In *Proceedings of IJCNLP*, pages 30–39, 2017.

[12] Katharina Hämmerl, Jindřich Libovickỳ, and Alexander Fraser. Understanding cross-lingual alignment—a survey. In *Findings of ACL*, pages 10922–10943, 2024.

[13] Ling Hu and Yuemei Xu. DM-BLI: Dynamic multiple subspaces alignment for unsupervised bilingual lexicon induction. In *Proceedings of ACL*, pages 2041–2052, 2024.

[14] Masoud Jalili Sabet, Philipp Dufter, François Yvon, and Hinrich Schütze. Simalign: High quality word alignments without parallel training data using static and contextualized embeddings. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 1627–1643, 2020.

[15] Judith F. Kroll and Erika Stewart. Category interference in translation and picture naming: Evidence for asymmetric connections between bilingual memory representations. *Journal of Memory and Language*, 33(2): 149–174, 1994.

[16] Guillaume Lample, Alexis Conneau, Marc'Aurelio Ranzato, Ludovic Denoyer, and Hervé Jégou. Word translation without parallel data. In *International Conference on Learning Representations*, 2018.

[17] Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, and Luke Zettlemoyer. BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7871–7880, 2020.

[18] Ping Li, Jennifer Legault, and Kaitlyn A. Litcofsky. Neuroplasticity as a function of second language learning: Anatomical changes in the human brain. *Cortex*, 58:301–324, 2014.

[19] Xintong Li, Guanlin Li, Lemao Liu, Max Meng, and Shuming Shi. On the word alignment from neural machine translation. In *Proceedings of ACL*, pages 1293–1303, 2019.

[20] Yinhan Liu, Jiatao Gu, Naman Goyal, Xian Li, Sergey Edunov, Marjan Ghazvininejad, Mike Lewis, and Luke Zettlemoyer. Multilingual denoising pre-training for neural machine translation. *Transactions of the Association for Computational Linguistics*, 8:726–742, 2020.

[21] Matthew W. Lowder and Fernanda Ferreira. Prediction in the processing of repair disfluencies. *Language, Cognition and Neuroscience*, 31(1):73–79, 2016.

[22] Chunpeng Ma, Akihiro Tamura, Masao Utiyama, Tiejun Zhao, and Eiichiro Sumita. Encoder-decoder attention $\neq$ word alignment: Axiomatic method of learning word alignments for neural machine translation. *Journal of Natural Language Processing*, 27(3):531–552, 2020.

[23] Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S. Corrado, and Jeff Dean. Distributed representations of words and phrases and their compositionality. *Advances in Neural Information Processing Systems*, 26, 2013.

[24] Tomáš Mikolov, Wen-tau Yih, and Geoffrey Zweig. Linguistic regularities in continuous space word representations. In *Proceedings of NAACL-HLT*, pages 746–751, 2013.

[25] Rico Sennrich and Biao Zhang. Revisiting low-resource neural machine translation: A case study. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 211–221, 2019.

[26] Rico Sennrich, Barry Haddow, and Alexandra Birch. Improving neural machine translation models with monolingual data. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics*, pages 86–96, 2016.

[27] Freda Shi, Luke Zettlemoyer, and Sida I. Wang. Bilingual lexicon induction via unsupervised bitext construction and word alignment. In *Proceedings of ACL-IJCNLP*, pages 813–826, 2021.

[28] Kaitao Song, Xu Tan, Tao Qin, Jianfeng Lu, and Tie-Yan Liu. MASS: Masked sequence to sequence pre-training for language generation. In *Proceedings of the 36th International Conference on Machine Learning*, pages 5926–5936, 2019.

[29] Yuqing Tang, Chau Tran, Xian Li, Peng-Jen Chen, Naman Goyal, Vishrav Chaudhary, Jiatao Gu, and Angela Fan. Multilingual translation with extensible multilingual pretraining and finetuning. *arXiv preprint arXiv:2008.00401*, 2020.

[30] Qiyu Wu, Masaaki Nagata, Zhongtao Miao, and Yoshimasa Tsuruoka. Word alignment as preference for machine translation. In *Proceedings of EMNLP*, pages 3223–3239, 2024.

[31] Thomas Zenkel, Joern Wuebker, and John DeNero. End-to-end neural word alignment outperforms GIZA++. In *Proceedings of ACL*, pages 1605–1617, 2020.